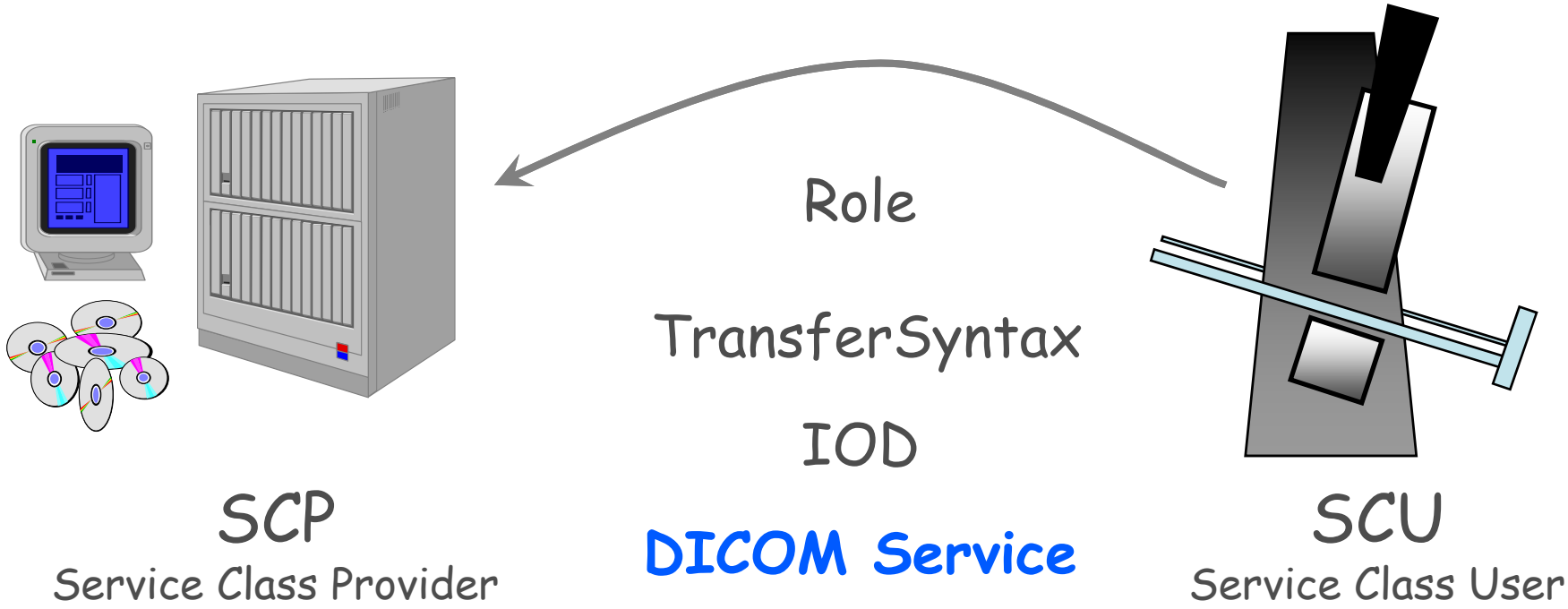
PHILIPS

DICOM Network Service

Agenda

- DICOM Network Service
  - Command Set
  - Data Set
- Composite Services
  - Storage
  - Query/Retrieve
- Normalized Services
  - Storage Commit
  - Print

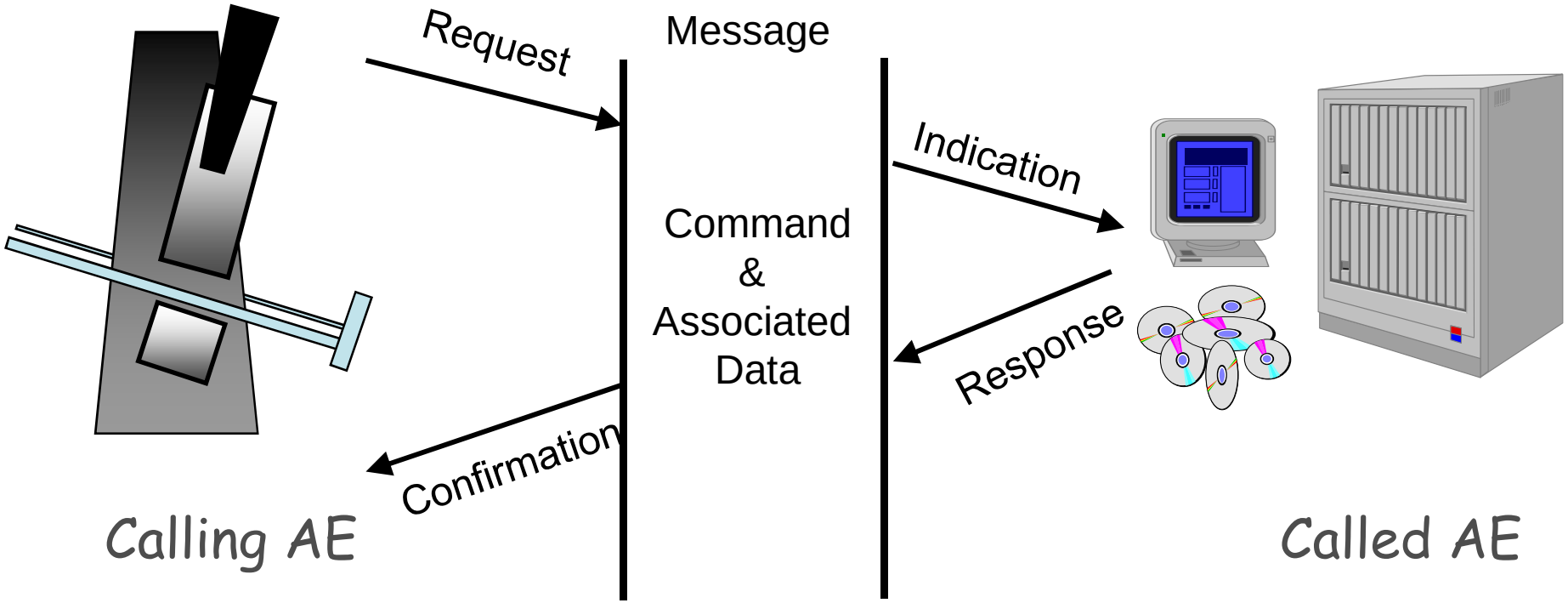
DICOM Network Model



DICOM Network Service

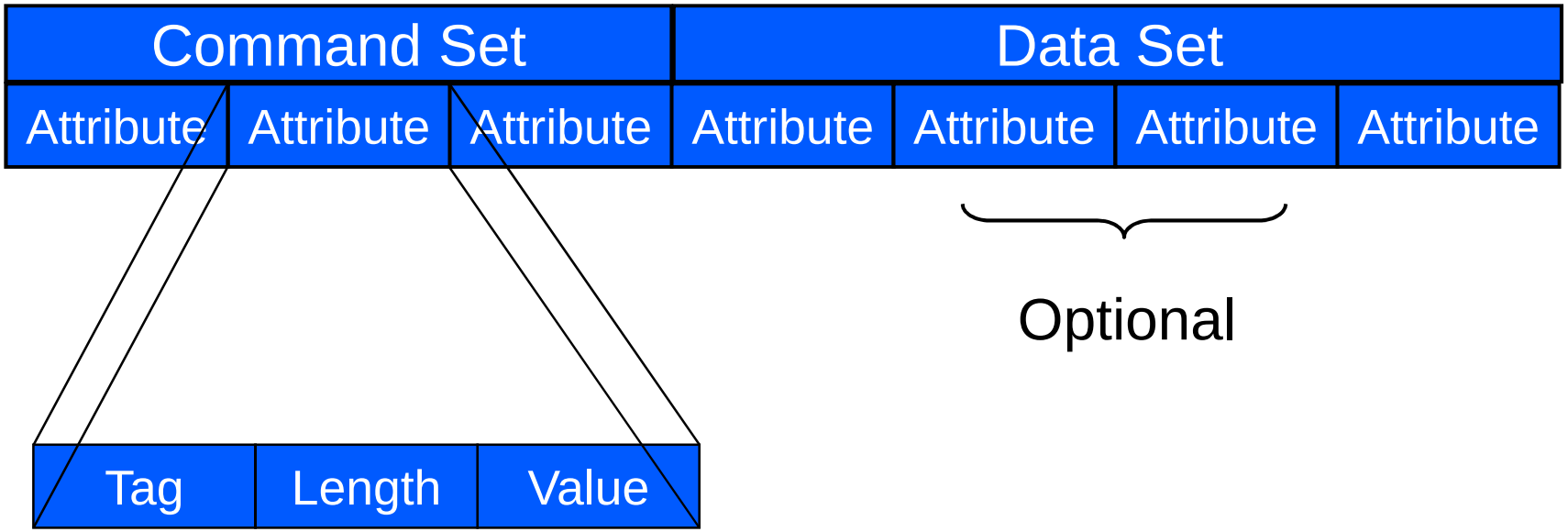
- DICOM Network Service begins after a Successful Association
- Encoded using P-DATA-TF PDUs
- Handled via SOP Classes
- SOP Class consists of
  - Service
    - Also known as DICOM Message Service Element (DIMSE)
  - Information Object Definition (IOD)

DICOM Message Service



DICOM Message Service

- DICOM Message Consists of
  - Command Set
  - Data Set



Command Set

- Indicates
  - Command being executed (Query/Storage etc…)
  - Presence of Optional Data Set
- Always encoded in Implicit VR Little Endian
- Request Command Contains
  - Message ID
- Response Command Contains
  - Message ID Being Responded To
  - Success/Failure of the Service
  - Optional Reason for Failure

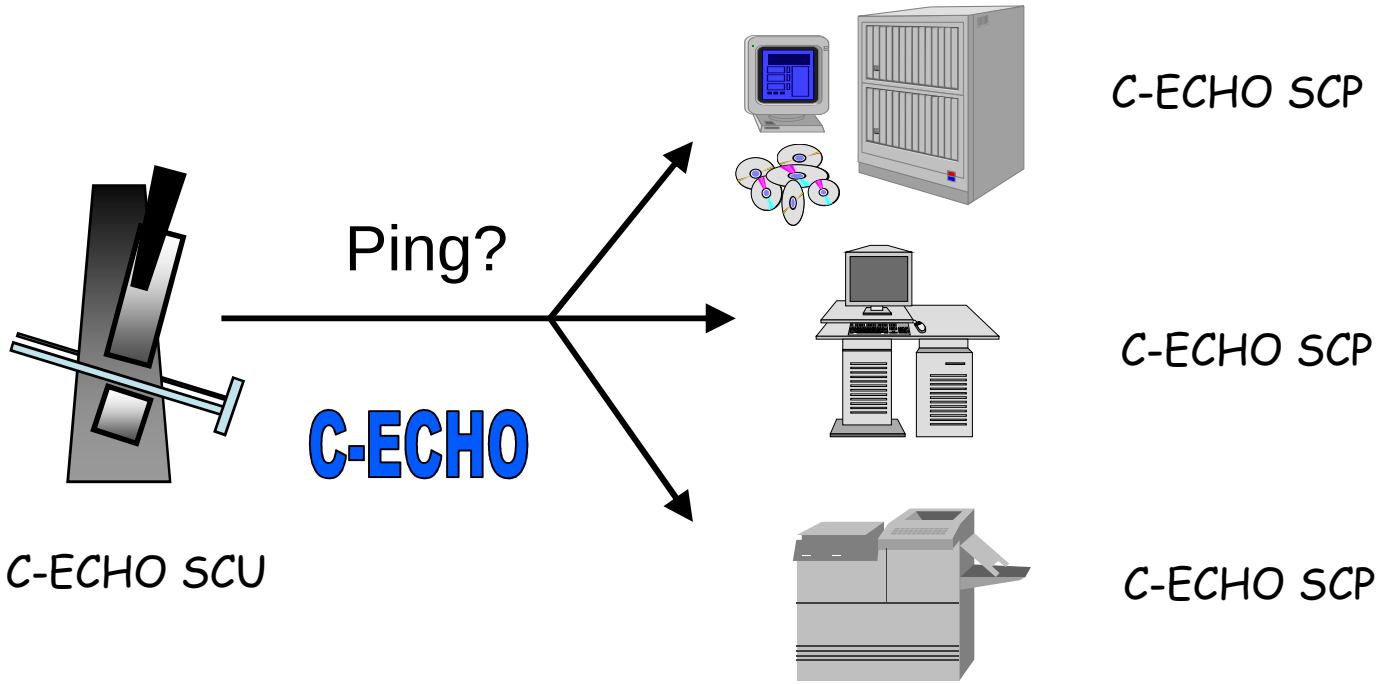
DICOM Network Service

- Two Types of Network Services
  - Composite Services
    - Services Related to Composite IODs
      - Verification
      - Storage
      - Query/Retrieve
      - Modality Worklist
  - Normalized Services
    - Services Related to Normalized IODs
      - Storage Commitment
      - Print Management

Composite Services

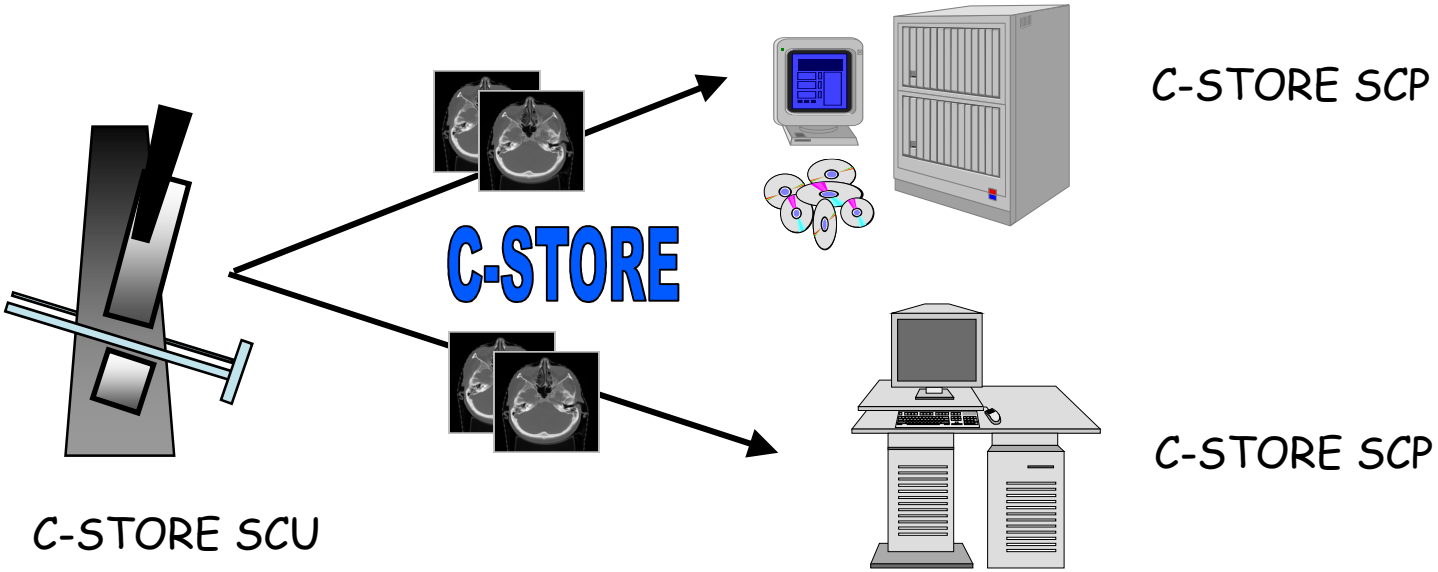
Verification Service

- Verifies the Availability of the Remote Node
- Implemented using the command DIMSE **C-ECHO**
- Does not contain a Data Set



Storage Service

- Transfers Composite Instances to a Remote System
- Used by Modalities/Work Stations to Save Images on PACS
- Implemented using the command DIMSE **C-STORE**

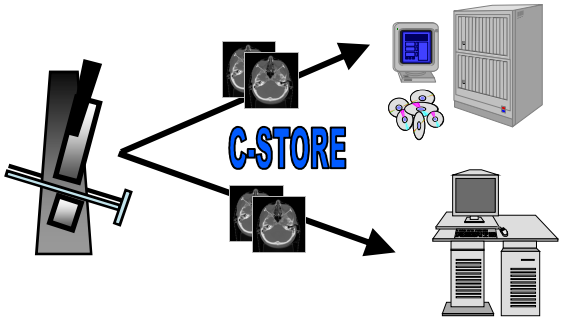


Storage Service

C-STORE Command Parameters

DIMSE-C Parameter Name	Req/Ind	Rsp/Conf
Message ID	M	—
Message ID Being Responded To	—	M
Affected SOP Class UID	M	U(=)
Affected SOP Instance UID	M	U(=)
Priority	M	—
Move Originator Application Entity Title	U	—
Move Originator Message ID	U	—
Data Set	M	—
Status	—	M

Table from DICOM 3.7.9.1.1

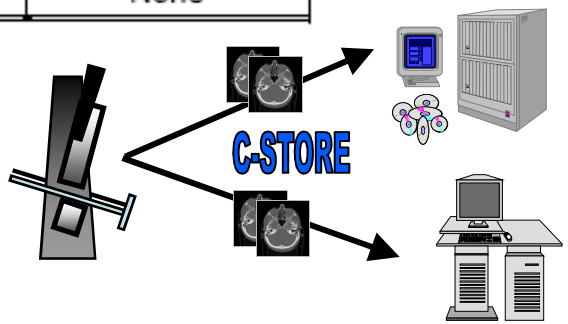


Storage Service

C-STORE Status Codes

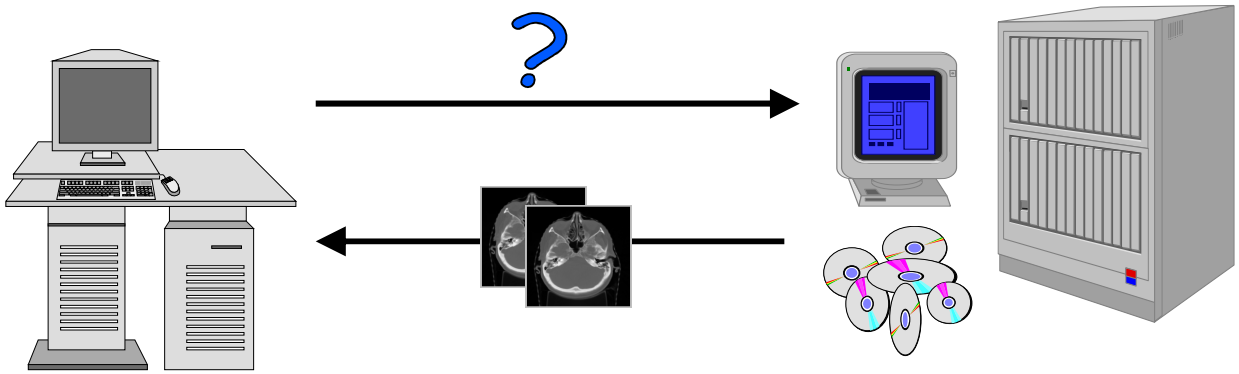
Service Status	Further Meaning	Status Codes	Related Fields
Failure	Refused: Out of Resources	A7xx	(0000,0902)
	Error: Data Set does not match SOP Class	A9xx	(0000,0901) (0000,0902)
	Error: Cannot understand	Cxxx	(0000,0901) (0000,0902)
Warning	Coercion of Data Elements	B000	(0000,0901) (0000,0902)
	Data Set does not match SOP Class	B007	(0000,0901) (0000,0902)
	Elements Discarded	B006	(0000,0901) (0000,0902)
Success		0000	None

Table from DICOM 3.4.B.2.3



Query/Retrieve Service

- Used For
  - Locating the Objects on a Remote System
  - Retrieving the Relevant objects from a Remote System
- Used by Work Stations to Retrieve the Objects from PACS/Modalities
- Implemented by DIMSE **C-FIND & C-MOVE**

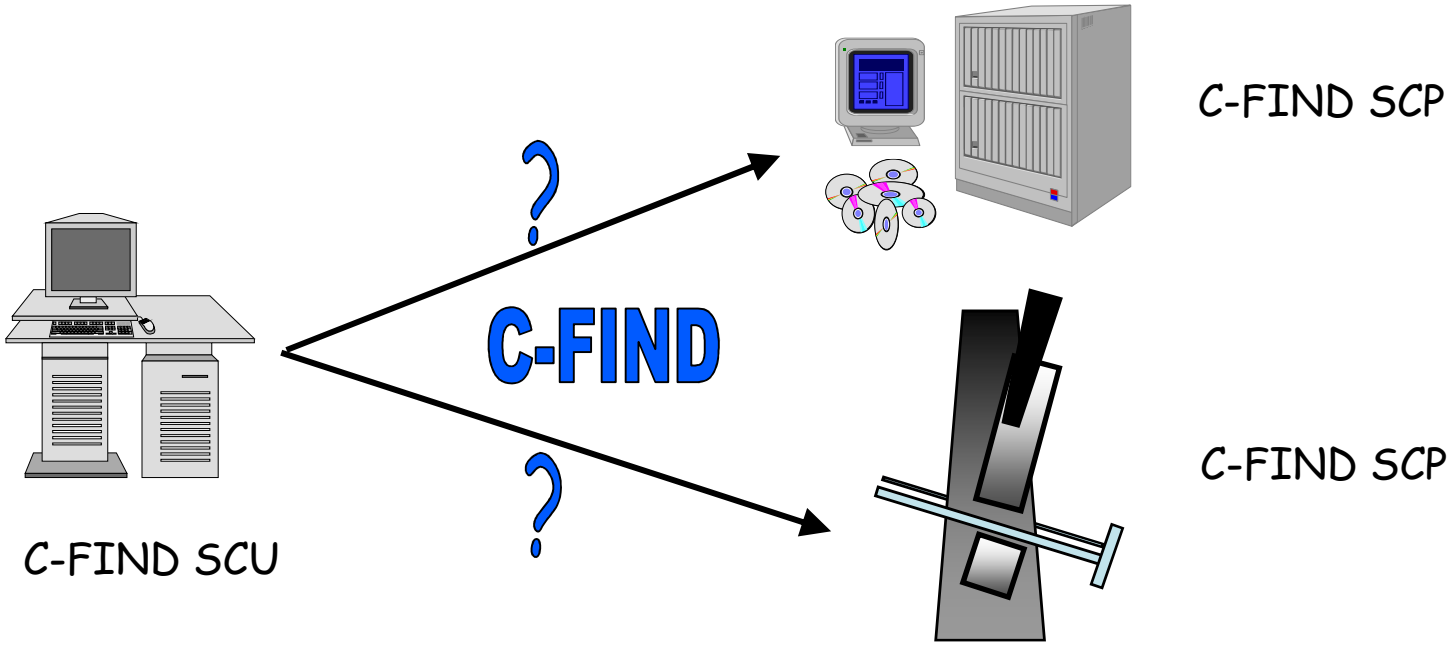


Query/Retrieve Information Model

- Hierarchical in Nature
- Patient Root Model
  - Patient, Study, Series and Image Level Hierarchy
- Study Root Model
  - Study, Series and Image Level Hierarchy

Query/Retrieve : C-FIND

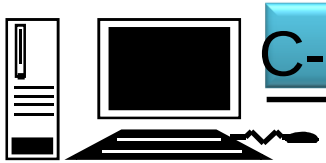
- Query the Remote System for Composite Objects
- Used by Work Stations to Query for the Objects on PACS/Modalities



Query/Retrieve : C-FIND

- Attribute Matching
  - Single Value Matching
    e.g: Patient ID == “1234”
  - Wild Card Matching
    e.g: Patient Name == “A*”, Patient ID == “123?”
  - Universal Matching
    e.g: Patient Name == “” / “*”
  - Range Matching
    e.g: Study Date >= “20070701”
    Study Time <= “153020”
    Study Date >=“20070701” &
    Study Date <= “20070723”

Query Remote Device

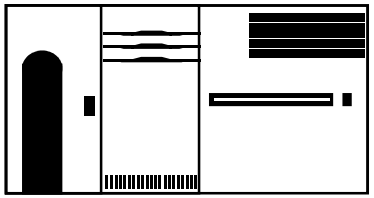


Workstation

Query SCU

C-FIND-RQ

-C-FIND-RQ Command
- Study=today, give me name/ID, etc.



Shared Archive

Query SCP

C-FIND-RSP

-C-FIND-RSP Command
(Pending Status)
- Study=Today, Joe Smith, 13456-7

C-FIND-RSP

-C-FIND-RSP Command
(Pending Status)
- Study=Today, Phil Jones, 2761-2

C-FIND-RSP

-C-FIND-RSP Command
(Success Status)

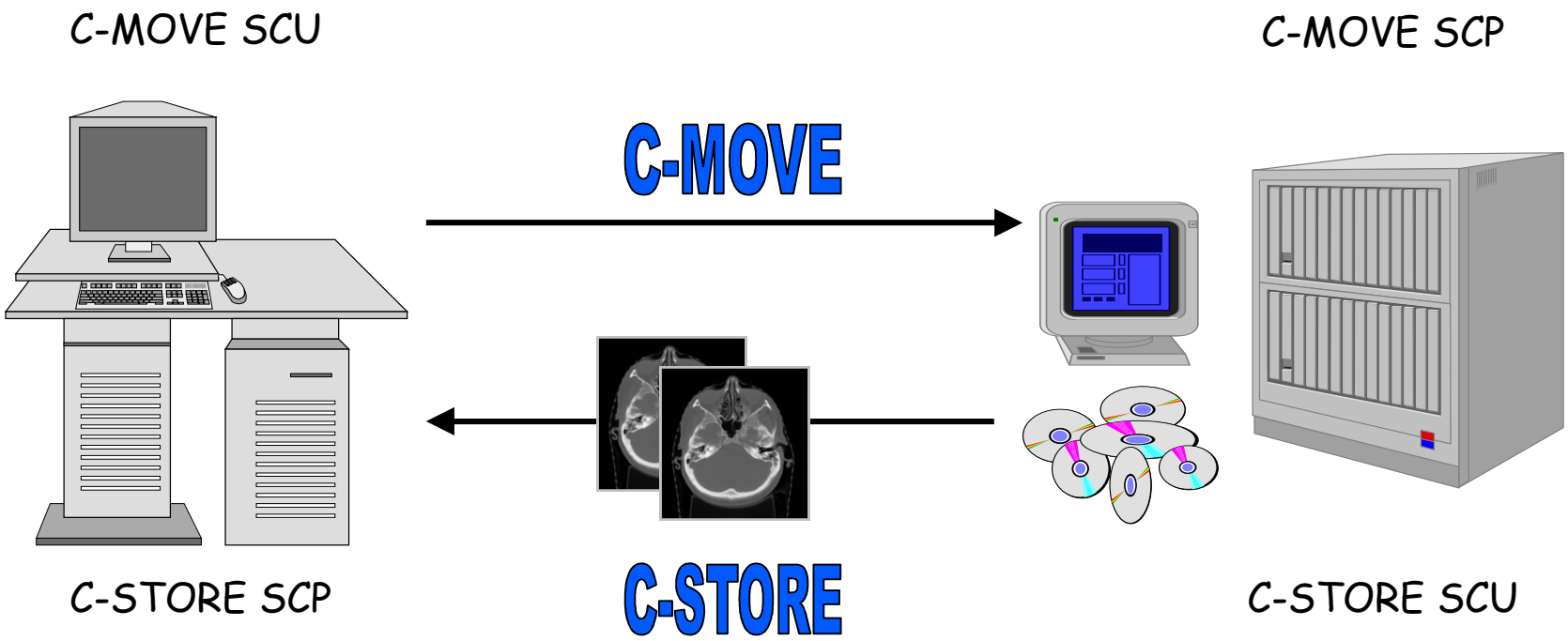
- Query remote database
- Pending rsp for each match
- One rsp for final status

Query/Retrieve : C-MOVE

- Transfers Composite Objects from One Remote System to another
- Used by Work Stations to Retrieve the Objects from PACS/Modalities
- A C-MOVE always results in a C-STORE
- C-MOVE always results in 2 DICOM Associations
  - One for C-MOVE
  - One for C-STORE
- C-MOVE can be realized as Two Ways

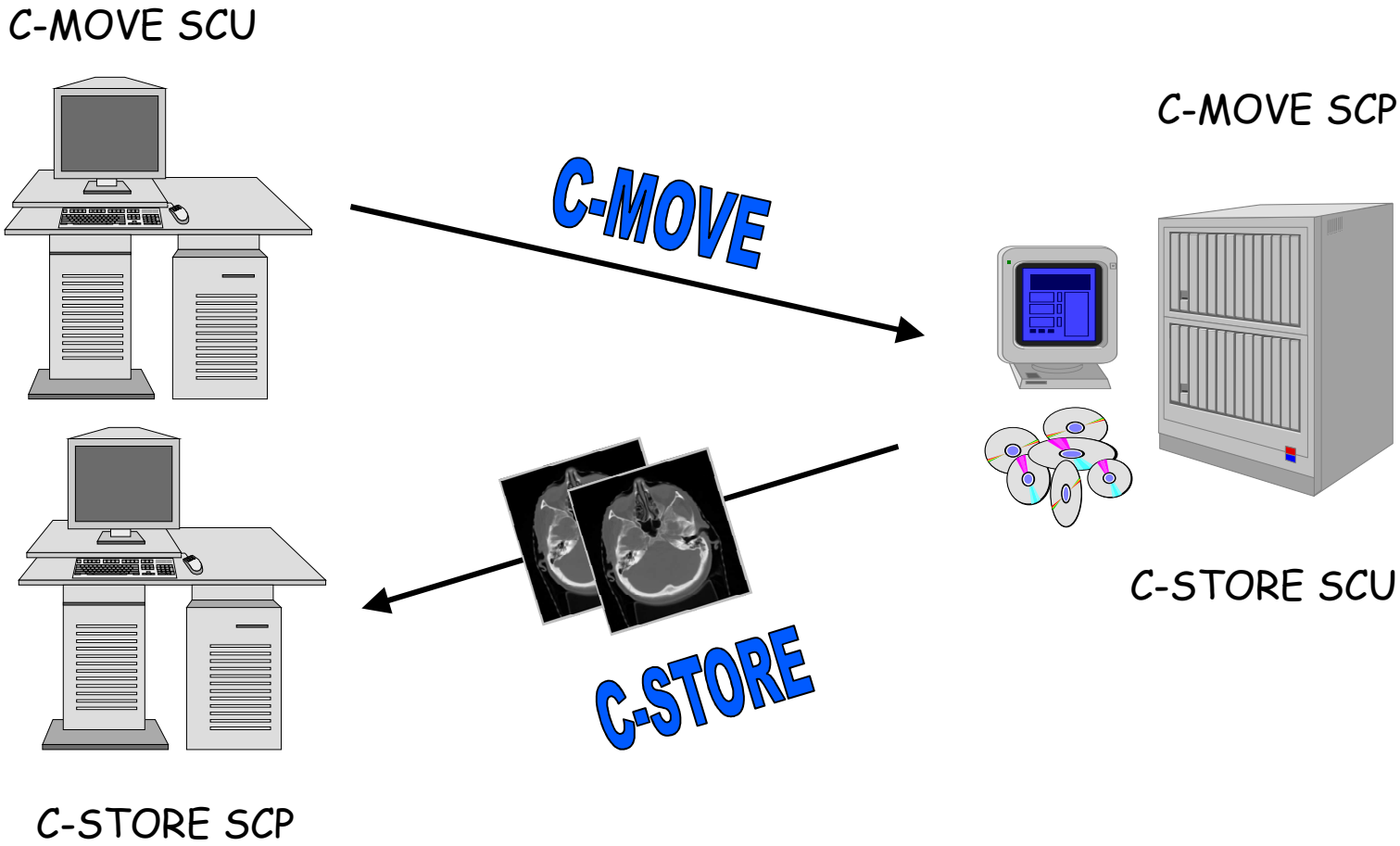
Query/Retrieve : C-MOVE

1. A Workstation Retrieving the Images from PACS



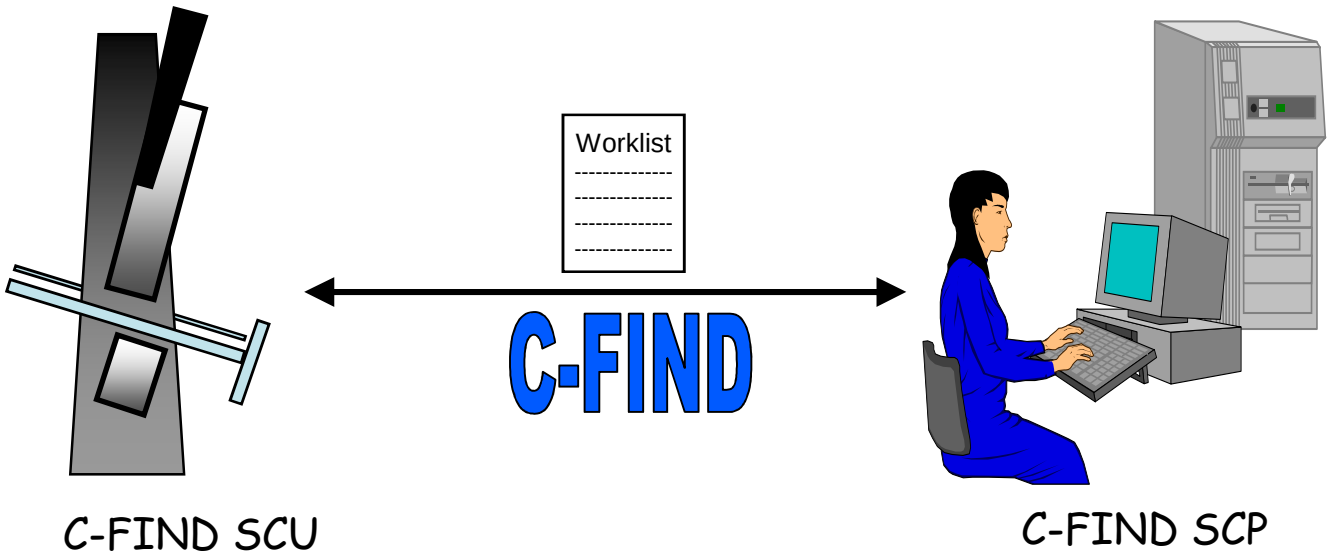
Query/Retrieve : C-MOVE

2. A Workstation Transferring the Images from PACS To another Workstation



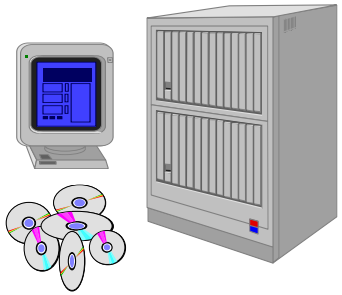
Modality Worklist : C-FIND

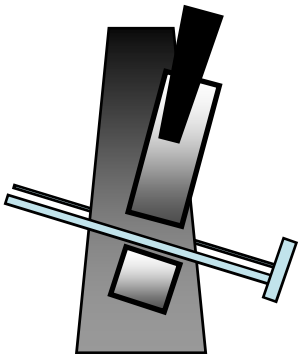
- Query the RIS/CIS for the Scheduled Patients
- Used by Modalities to fetch Patient/Study Information
- Similar to the Q/R C-FIND except for the Information Model



Exercises

Identify the Composite Services possible





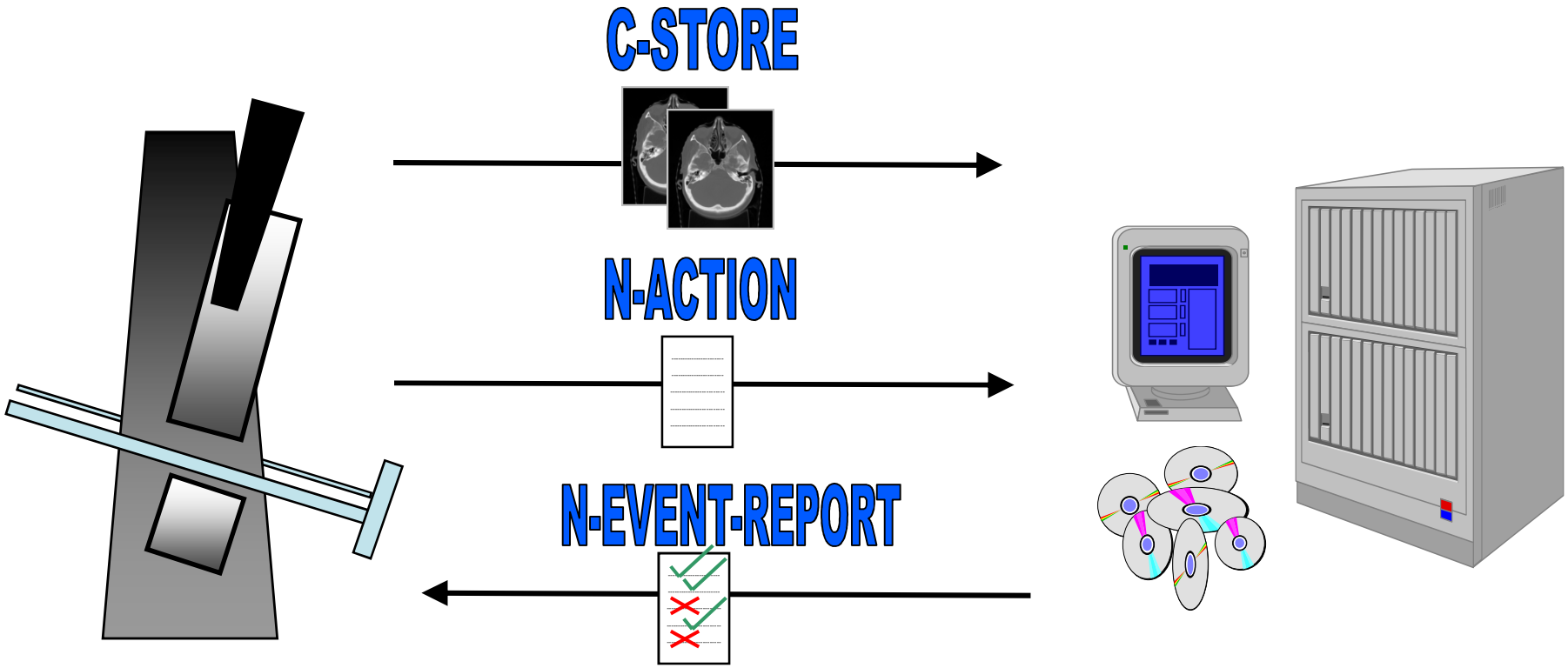
Normalized Services

Storage Commitment

- Storage Service
  - Allows SCU to Transmit Images to SCP
  - SCP may delete the Data
  - Does not explicitly ensure the Safekeeping of received Data
- Need for Explicit Commitment of the Stored Objects
- Storage Commitment
  - Transfers the Responsibility of Objects to the SCP
  - SCP Implementation defines the Safekeeping Policy
  - Used by Modalities to make sure that they can delete the Composite Objects

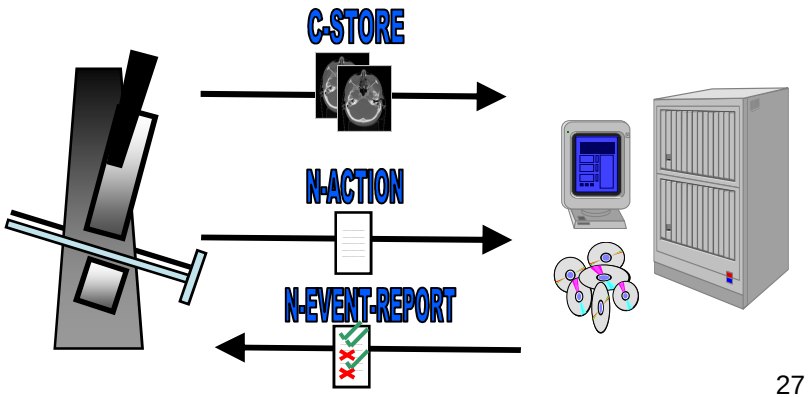
Storage Commitment

- Implemented using the Storage Commit SOP Class
- Implemented by DIMSE **N-ACTION & N-EVENT-REPORT**



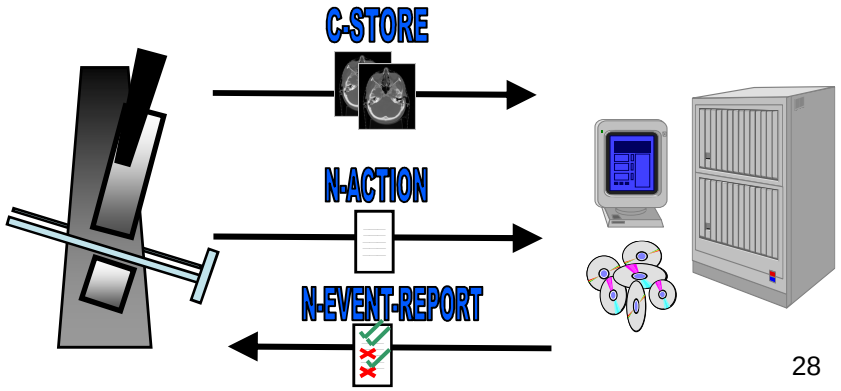
Storage Commitment

- Two Ways of Implementing Storage Commitment
  - Synchronous Storage Commitment
    - SCP Sends the N-EVENT-REPORT Immediately in the same Association
  - Asynchronous Storage Commitment
    - SCP Sends the N-EVENT-REPORT later in a separate Association



Storage Commitment

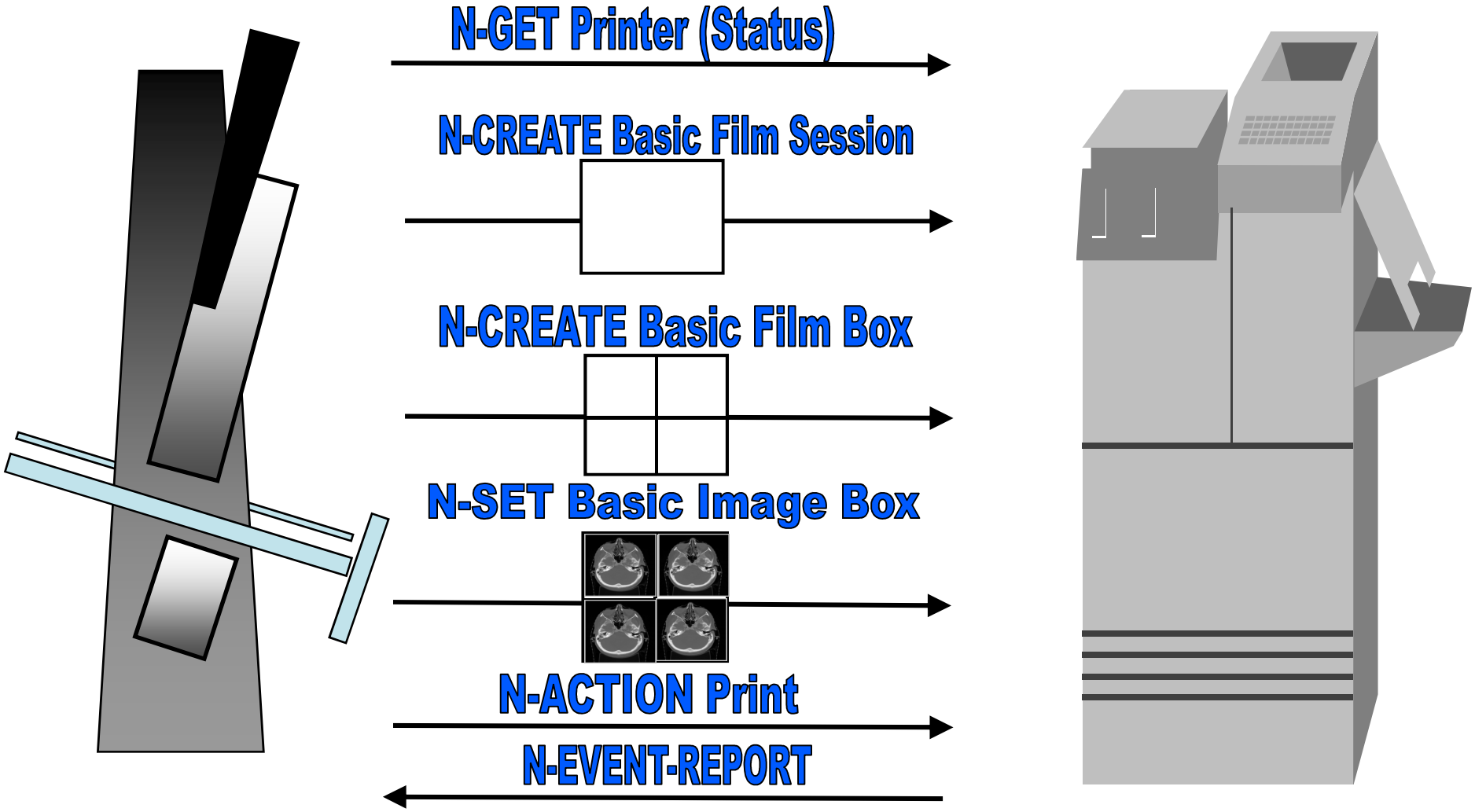
- Storage Commit Request (N-ACTION) Contains
  - Transaction UID
  - List of SOP Class UID & SOP Instance UID
- Storage Commit Response (N-EVENT-REPORT) contains
  - Transaction UID
  - Successful SOP Class UID & SOP Instance UID
  - Failed SOP Class UID & SOP Instance UID and Reason for Failure



Print Management

- Prints the Images on a Film
- Implemented by Basic Grayscale/Color Print Management Meta SOP Class
  - Meta SOP Class is used only for Association
  - Needs Four SOP Classes to perform the Print
    - Basic Film Session SOP Class
    - Basic Film Box SOP Class
    - Basic Grayscale/Color Image Box SOP Class
    - Printer SOP Class
  - Optionally Printer Job SOP Class

Print Management Sequence



Print Management

- DIMSE Service Used
  - N-ACTION
    - Print the Film Session
  - N-CREATE
    - Create Film Session/Film Box
  - N-DELETE
    - Delete Film Session/Film Box
  - N-GET
    - Get the Status of the Printer and the Print Job
  - N-SET
    - Set the Image Box inside Film Box
  - N-EVENT-REPORT
    - Report Printer Status/Print Status

Exercise

- Identify the following in the DICOM Statements below
  - Roles SCU/SCP
  - Called AE Title/Calling AE title
  - Service Class UID Used
  - Transfer Syntax UID
  - IOD

  – The AE Titles used are:

    ViewForum = VF MR Modality = MR XCelera = XC

MR Modality Stores an MR Image on XCelera in JPEG Lossless Format.

ViewForum Queries the XCelera for Patients in ImplicitVR LittleEndian

ViewForum Retrieves an RLE US Image from XCelera in ExplicitVR BigEndian

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sense and simplicity